Causal Regime-Conditioned Trading Agent: Verifiable Regime-Aware Reinforcement Learning

A Project Synopsis submitted to the
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1. INTRODUCTION

Cryptocurrency markets are highly volatile and non-stationary. Within a single multi-year cycle, the same asset may experience strong upward trends, sharp crashes, low-volatility consolidation, and liquidity crises. A reinforcement learning (RL) agent trained without representing the prevailing market state may learn an averaged policy that generalises poorly when market conditions change.

Recent advances in deep reinforcement learning, regime-switching time-series models, and explainable artificial intelligence (XAI) provide a way to build trading policies that adapt to detected market regimes. Hidden Markov Models (HMMs) have been used to infer latent market states, while risk-adjusted deep RL frameworks have incorporated multiple reward objectives to produce more risk-aware trading behaviour.

However, a major limitation in existing regime-aware trading research is that the influence of the regime signal on the learned policy is often assumed rather than demonstrated. Research on XAI in finance has also identified a lack of standardised evaluation metrics for explainability. This project therefore proposes a regime-conditioned cryptocurrency trading agent together with a causal verification framework to test whether the regime signal actually influences trading decisions.

2. LITERATURE SURVEY

Koukorinis, Peters and Germano proposed a hybrid generative-discriminative approach for high-frequency financial regime classification. Their method uses a Hidden Markov Model to generate latent market-state embeddings, which are then supplied to support vector machine classifiers. The study identifies multiple intraday regimes and reports improved classification performance compared with several conventional approaches. It demonstrates that latent-state models can capture market regimes that may be missed by traditional time-series methods.

However, the study evaluates the regime label mainly as a classification output and does not investigate whether the detected regime changes the behaviour of a downstream trading policy.

Cerneviciene and Kabasinskas conducted a systematic review of 138 studies on explainable AI in finance. They found that feature importance, SHAP-based methods, and rule-based approaches are among the most commonly used explanation techniques. The review highlights the lack of reliable evaluation metrics for explainability and stresses that explanations should be faithful to the underlying model. It also shows that most existing work focuses on static supervised prediction rather than sequential decision-making agents, leaving a gap in verifying explanations for RL-based trading systems.

Choudhary, Orra, Sahoo and Thakur proposed a risk-adjusted deep RL framework for portfolio optimisation. They trained multiple PPO agents using different reward functions based on returns, differential Sharpe ratio and maximum drawdown, and combined their actions into a unified policy. Their results demonstrate that reward design can improve risk-aware behaviour. However, the framework does not explicitly provide the prevailing market regime to the agent and does not verify which inputs are responsible for its allocation decisions.

3. PROBLEM STATEMENT

RL trading agents are generally trained on mixed historical market conditions and may therefore learn behaviour averaged across economically different regimes. Even when a regime signal is provided, evaluation is commonly based on aggregate profitability, which cannot determine whether the agent genuinely uses the regime information or simply ignores it.

Evaluation can also be affected by look-ahead leakage, inappropriate train-test splits, and single-point performance measurements. Therefore, there is a need for a systematic method to verify whether a regime signal is actually influential in a trading policy.

4. OBJECTIVES

The proposed system aims to:

1. Build a reproducible cryptocurrency data and feature pipeline using strict no-look-ahead constraints.
2. Detect market regimes using a sticky Hidden Markov Model and generate both regime labels and confidence values.
3. Train a regime-blind baseline agent and a regime-conditioned agent using identical architectures and training budgets.
4. Develop a causal verification framework using ablation, counterfactual, and sensitivity tests.
5. Evaluate the agents using walk-forward and regime-segmented backtesting with bootstrapped confidence intervals.

5. PROPOSED METHODOLOGY

The proposed system collects historical cryptocurrency OHLCV and order-book data and generates features such as returns, rolling volatility, moving-average crossovers, RSI, and MACD while preventing look-ahead leakage.

These features are provided to a Gaussian Hidden Markov Model with a minimum dwell-time constraint to identify four market regimes: bull, bear, calm, and crisis, along with their confidence values. The regime information, market state, and portfolio state are then provided to a regime-conditioned reinforcement learning agent, while a regime-blind agent with the same architecture and training settings serves as the baseline.

The conditioned agent is evaluated using input ablation, counterfactual injection, and sensitivity analysis to verify the influence of regime information. Finally, both agents undergo walk-forward and regime-wise backtesting using Sharpe ratio, Sortino ratio, drawdown, and bootstrapped confidence intervals to compare performance and policy behaviour.

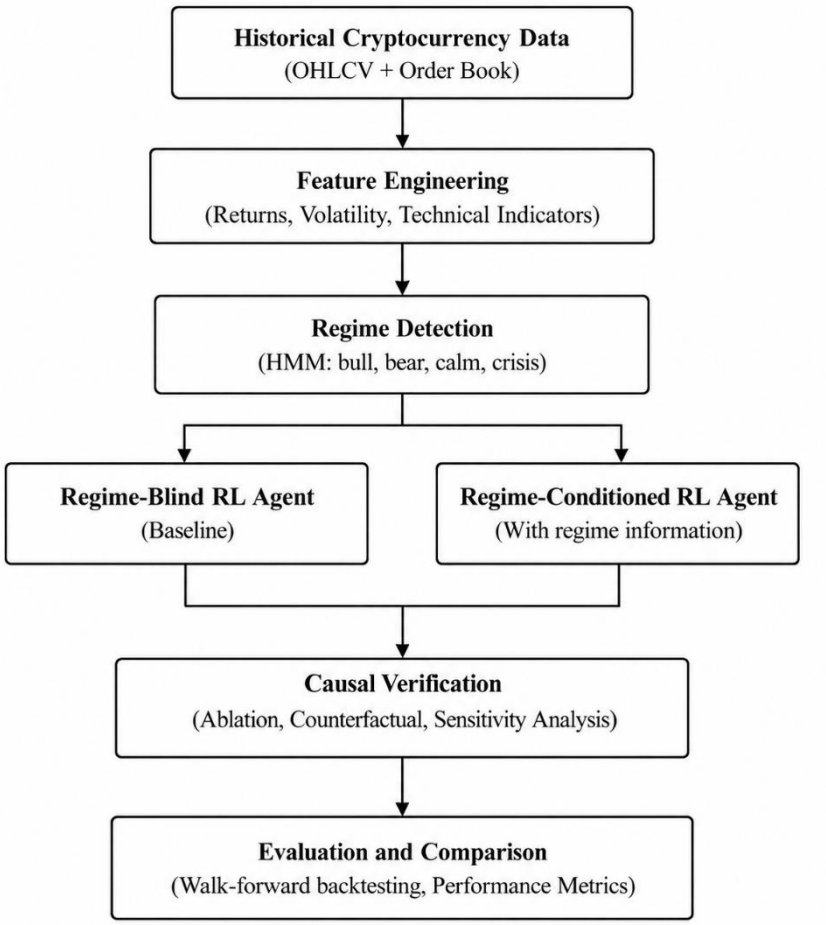


Figure 1: Proposed System Workflow

6.REFERENCES

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